What a Campus Network Actually Looks Like: Traffic Composition, Concentration and Short-Window Stability

ST5290 Data Science Industry Project · Zhejiang SuanWangYun Intelligent Technology Co., Ltd.

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Abstract

This report analyses one packet capture taken at the core collection point of an enterprise campus network: four files, 36.2 GB and 68.7 million packets, covering an effective window of two continuous blocks totalling 17.2 minutes, with 49,560 hosts (5,223 of them inside) and 693,484 sessions. The analysis proceeds in four steps, namely applying a one-way byte convention, building host, session and name views, running four detections on periodicity, port-entropy bimodality, change points and extremes, and turning the results into findings and recommendations. The key findings are:

- **Receive-dominant**: inbound bytes are 3.64 times outbound while packet counts are almost equal in both directions.
- **Highly head-heavy**: the host-level Gini coefficient is 0.982 and the top 1% of hosts carry 81.6% of the bytes.
- **Four host roles**: serving, interactive, bulk and low-volume, of which 478 serving hosts carry 72.2% of the inside bytes.
- **Encrypted service face outward, service mesh inward**: outbound packets on port 443 run 5,317 times inbound.

The four detections produced ten candidates, none of which satisfies all three conditions required for promotion to a confirmed anomaly. The most important limitation is the 17.2-minute window, so the structural findings can be quoted while the time-based ones hold only inside it. Overall, **this is a highly head-heavy, receive-dominant campus network**, an almost purely outbound encrypted service face from the outside and a service mesh on the inside.

1 Introduction

1.1 Background and related work

Describing the traffic of enterprise and campus networks from packet captures is a long-established line of work. Early studies established two facts that still shape the field: traffic is extremely unevenly distributed across hosts and flows, with a few long-lived connections carrying most of the bytes, which Brownlee and Claffy named dragonflies and tortoises [1]; and hosts in different roles differ sharply in their traffic features, so managing by group works better than managing by network-wide average. Data-centre studies quantified this unevenness and showed that bottlenecks concentrate on a few racks or peers [2], [3]; the campus network analysed here shows the same shape.

Encryption has changed what can be observed. Once payloads are protected by TLS, content-based identification largely fails, so analysis works on metadata: arrival times, packet sizes, byte counts, ports and external identifiers such as DNS names and TLS server names. This limits which questions can be answered, since business content is out of reach, and makes reasoning about statistical shape rather than content a necessity.

This project relates to that work in four ways:

- **Inside and outside in one dataset**: inside roles and outward identity can be compared within one dataset rather than by joining two differently defined sources.
- **Window length treated as a limitation**: the real shape of the window (two continuous blocks, 17.2 minutes) is stated before every time-based claim is limited to it.
- **Leads separated from anomalies**: promotion conditions are explicit and auditable.
- **A transparent rule**: the grouping rule can be recomputed by hand rather than being a black-box output that has to be accepted.

1.2 Project origin and research questions

This project was carried out during an internship at Zhejiang SuanWangYun Intelligent Technology Co., Ltd. The object of analysis is the complete traffic record of one capture taken at the core collection point of the company's campus network, whose scale and conventions are described in Section 2. The topic was chosen because this capture contains both the communication among hosts inside the campus and their communication with external peers, so one dataset can answer what

the network looks like on the inside and how it behaves towards the outside, without reconciling two differently defined sources.

The report is organised around three research questions:

- **Network structure**: what the structure of this network is, which includes who talks, how concentrated the traffic is, and in which direction it flows.
- **Stability in time**: whether the network is stable over time, which includes whether the window shows a trend, a periodic rhythm, or an abrupt change.
- **Leads worth following**: whether the data contains leads worth following, that is, which hosts or behaviours deviate substantially from the group they belong to.

2 Data and environment

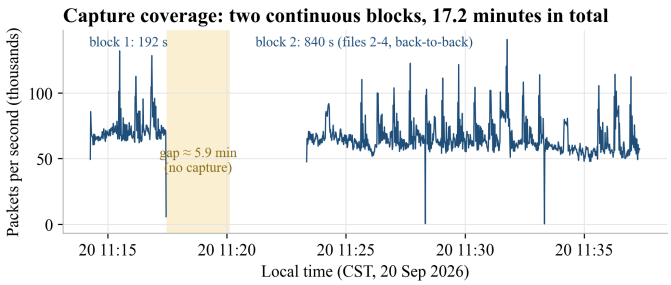
The raw data consists of one capture taken at the core collection point of the campus network, whose scale is summarised in the table below.

Item	Value	Note
Capture files	4	11:14:16 to 11:37:21 on the same morning
Effective window	**two continuous blocks**	192 s + 842 s, separated by 353 s (about 5.9 min), 1,034 s = 17.2 min in total
Raw volume	36.2 GB / 68.7 M packets	one-way traffic **34.85 GB**
Hosts	**49,560**	5,223 inside + 44,335 external + 2 loopback
Capture interfaces	27	the top three carry most traffic
Protocol mix	TCP about 81%, UDP about 9%, non-IP L2 about 9%	IPv6 only 0.003% (2,086 packets)

Three conventions determine how the numbers should be read.

- **One-way counting**: all traffic is measured in one-way bytes, counting every packet once, so the gigabyte figures here cannot be compared directly with the bidirectional counters shown on network devices. Of the 34.85 GB total, 34.26 GB falls into the five direction classes (the denominator used in Figure 3) and the remaining 0.59 GB (1.7%) is non-IP or unclassified traffic.
- **Separately reported traffic**: loopback traffic and traffic from a host to its own services are reported separately, as is non-IP layer-2 traffic, so neither is mistaken for service traffic.
- **Cross-interface duplicates**: because the capture used several interfaces at once, a few frames were recorded twice; this duplication is below 0.1% of the volume and moves no conclusion.

The full convention note is kept in the project as the data-convention document.



Each point = one second of capture; gaps are seconds with no data. Files 2/3/4 meet end-to-start, so they form one 840 s block. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 1: Capture coverage: two continuous blocks, 17.2 minutes in total

How to read this figure: the horizontal axis is local time and the vertical axis is packets per second, with one point per second. The blank interval in the middle is not a drop in traffic; it is a period during which no capture took place, which is why the line is broken there rather than connected. What matters most is the real shape of the observation window: it consists of two continuous blocks of 192 and 842 seconds, 353 seconds apart, rather than four separate fragments, and this sets the scope within which every time-based statement later in the report is valid.

3 Methodology

The analysis consists of four steps, and every stage is reproducible from a script kept in the project.

Concretely, the four steps are:

1. **Data convention**: loopback traffic, non-IP traffic and cross-interface duplicates are removed as described in the previous section.
2. **Three views**: a host profile of 49,560 rows with packets, bytes, peers, ports, port entropy and mean packet size per host; a session view of 693,484 sessions keyed by bidirectional five-tuples; and a name view of 13,957 DNS query names and 6,875 TLS server names.
3. **Four detections**: periodicity together with the coefficient of variation, bimodality of port entropy, change points (judged by a permutation test on a step in the mean), and extreme values.
4. **Findings and recommendations**: the detection results are turned into quotable findings, each labelled with its own confidence and limitations.

The grouping rule needs to be stated explicitly, because it determines what every later statement about groups means. It uses two criteria: hosts whose bytes reach the inside P90 count as **bulk**, and hosts with at least 50 ports and at least 20 peers as **wide**, which cross to four groups, namely serving, interactive, bulk and low-volume. A hand-checkable rule is preferred over clustering for two reasons. Host bytes are distributed in an extremely head-heavy way, so a mean-based threshold would place almost every host on the same side whereas a quantile does not, and because anyone can recompute the rule from the host profile, the grouping becomes an object that can be checked rather than a black-box output. Its agreement with a data-driven clustering is tested in Section 4.3. Note that these three thresholds are analyst-selected quantiles and integer cut-offs rather than quantities derived automatically from the data, which is acknowledged as a limitation in Section 7.

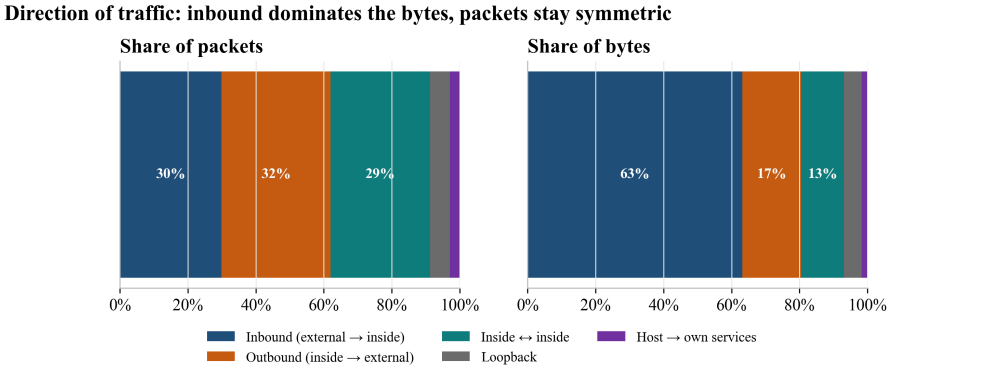
The data is used at two levels. The **deep pass** re-reads all four captures in full (68,696,735 packets) to obtain the second-level series, the complete session byte distribution and the host-level periodicity candidates, while the **sampled attribution** parses DNS names, TLS server names and device fingerprints in the first 500,000 packets of each file, two million packets in total, to infer the behavioural leaning of each segment. The sampled window is not random and covers only connections that asked for or presented a name, so it provides indicative rather than population evidence.

4 Results

4.1 Scale and traffic composition

The most basic observation concerns the shape of the traffic itself. The packet-length distribution is clearly bimodal: one mode sits between 64 and 127 bytes and corresponds to the large volume of control packets, while the other sits between 1024 and 1518 bytes and corresponds to fully loaded data packets. A further 14% of frames exceed 1518 bytes, which is beyond the Ethernet MTU and therefore originates from capture-side offload rather than from traffic on the wire. The mix of protocols and ports is comparatively concentrated: TCP carries about eighty percent of the packets and UDP about ten percent, with the remainder being non-IP layer-2 traffic, and on the destination side a single port, 443, accounts for a large share of all service-port packets, followed by port 80 and a small number of internal service ports.

The asymmetry of direction is the most important single observation in this report (see Figure 2). Inbound bytes amount to 21.63 GB, or 63.1% of the one-way total, while outbound bytes amount to 5.95 GB, or 17.4%, giving a ratio of **3.64x**. In sharp contrast, the packet counts in the two directions are almost equal, with a ratio of only 0.93x.



Byte shares, denominator = sum of the five direction classes (34.26 GB): Inbound 63.1% (21.6 GB) · Outbound 17.4% (5.9 GB) · Inside ↔ inside 12.5% (4.3 GB) · Loopback 5.3% (1.8 GB) · Host → own services 1.7% (0.6 GB). In/out byte ratio 3.64x while the packet ratio is 0.93x. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

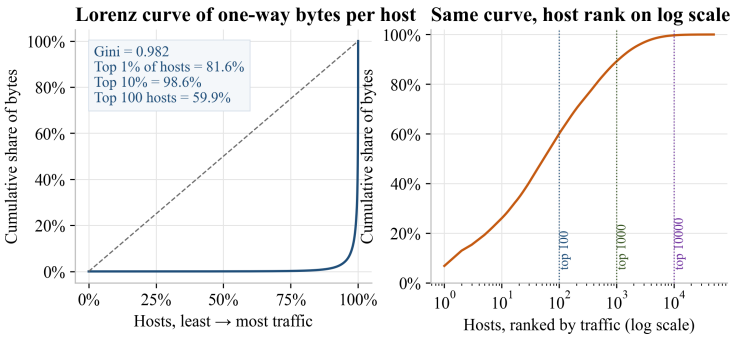
Figure 2: Direction of traffic: bytes come inbound while packets stay symmetric

How to read this figure: the two bands show the same directional breakdown, the upper one by packet count and the lower one by bytes, and they are meant to be compared with each other. The point to notice is that the two bands have completely different shapes: by packet count the two directions are nearly symmetric, whereas by bytes the inbound side is visibly wider.

This difference is why connection counts and data volumes have to be planned as two separate matters, and it is the direct basis for the capacity recommendation later in the report.

4.2 Concentration: averages fail here

The distribution of traffic across hosts is extremely uneven (see Figure 3). The figure shows this with a Lorenz curve, which plots the cumulative share of bytes against hosts ordered from the smallest to the largest volume, the most direct way to see whether tiered management is needed [4]. Measured in one-way bytes, the host-level Gini coefficient (a measure of inequality, 0 for perfect equality and 1 for full concentration) is **0.982** [5]; the top 1% of hosts carry 81.6% of the bytes, the top 10% carry 98.6%, and the top 100 hosts alone carry 59.9%. In other words, out of nearly fifty thousand hosts, only a very small cluster actually moves the data.

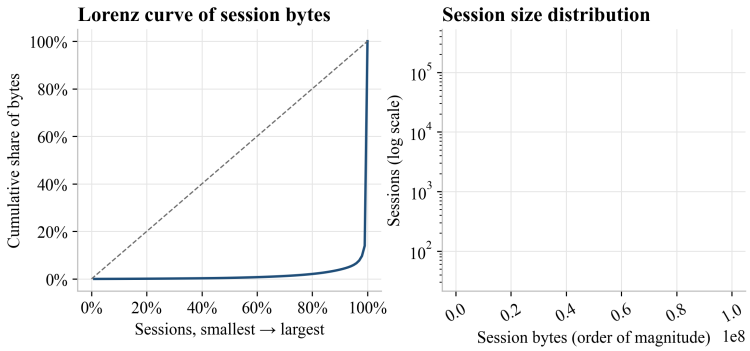


n = 49,560 hosts (5,223 inside + 44,335 external peers + 2 loopback), one-way bytes. Left tail of the log panel is where the capacity sits. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 3: Host concentration: the top 1% of hosts carry 81.6% of the bytes

How to read this figure: in the left panel the horizontal axis orders hosts from the smallest to the largest traffic volume and the vertical axis gives the cumulative share of bytes, with the diagonal dashed line representing perfect equality, so the closer the curve runs to the bottom-right corner the more uneven the distribution is. The right panel switches the horizontal axis to a logarithmic scale and accumulates from the largest host downwards, which shows where capacity is actually concentrated. The steepness at the left end of the right panel is the key detail, since the first hundred hosts already account for close to sixty percent of the bytes, and this directly determines how monitoring thresholds should be set.

The same concentration appears at the session level (see Figure 4). Across 693,484 sessions the Gini coefficient of session bytes is **0.97**; the ten largest account for 10.5% of all session bytes, the top hundred for 34.9% and the top thousand for 66.1%, while most sessions are very small: 330,000 sit in the 10^2 byte range and only 34 exceed 0.1 GB.



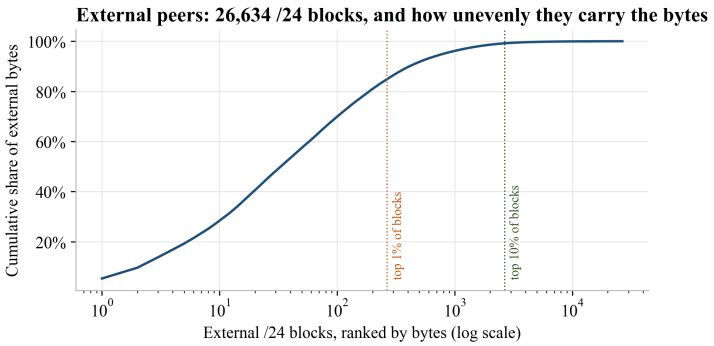
n = 693,484 sessions (≥2 packets); Gini = 0.970; top 10 = 10.5%, top 100 = 34.9%, top 1,000 = 66.1%. Session = bidirectional 5-tuple with ≥2 packets. Byte total 34.23 GB, i.e. 98.2% of the 34.85 GB one-way total; median duration 1.0 s, P90 90.0 s. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 4: Session concentration: the top 100 of 693,484 sessions carry 34.9% of the bytes

How to read this figure: the left panel is the Lorenz curve of session bytes and the right panel is the distribution of session sizes on a logarithmic vertical axis. What matters is the combination of the two: sessions are extremely numerous and almost all of them are small, yet the bytes are carried by a very small number of elephant flows. This explains why a capacity estimate based on session counts alone would substantially understate bandwidth pressure.

The same pattern holds when the view moves to external peers (see Figure 5). Among 26,634 candidate /24 blocks, the top 1% carry 85.0% of the external bytes and the top 10% carry 99.2%. The convention matters here: a /24 block can only be treated as an approximation of one external organisation, and it is neither a single host nor a person.

How to read this figure: the horizontal axis ranks external /24 blocks by bytes on a logarithmic scale and the vertical axis gives the cumulative share of bytes. The shape of the curve is the key point, because it completes most of its rise within



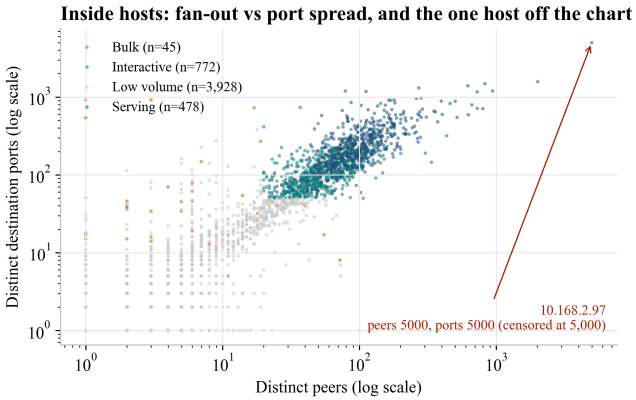
Top 1% of blocks carry 84.9% of external bytes, top 10% carry 99.2%. A /24 is an approximation of one external organisation, not a person. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 5: External peers at /24 granularity: the top 1% of blocks carry 85.0% of the bytes

a very short stretch at the left end, which means that external traffic is in practice concentrated among very few external organisations and answers directly the question of whose usage should drive outbound link capacity.

4.3 Group structure

Grouping hosts by behaviour makes the internal structure of the network clear (see Figure 6). The 5,223 inside hosts fall into four groups: 478 serving hosts, or 9.2% of the hosts, carry 72.2% of the inside bytes; 772 hosts are interactive; 45 are bulk transfer hosts; and 3,928 hosts, or 75.2%, are low-volume hosts that together account for only 0.8% of the bytes. Port entropy is clearly bimodal, with 52.4% of inside hosts using a single port while another set spreads across many ports, and this bimodal shape is itself the basis for splitting hosts into groups. As for the segments, the differences show up in **behaviour** rather than in **nature**: service-leaning segments have fewer peers and ports but a longer mean packet, yet without sampled evidence no segment should be labelled a server or business zone outright.



n = 5,223 inside hosts. Groups follow the same transparent rule as the report (bulk = bytes ≥ P90; wide = ports ≥ 50 and peers ≥ 20). Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

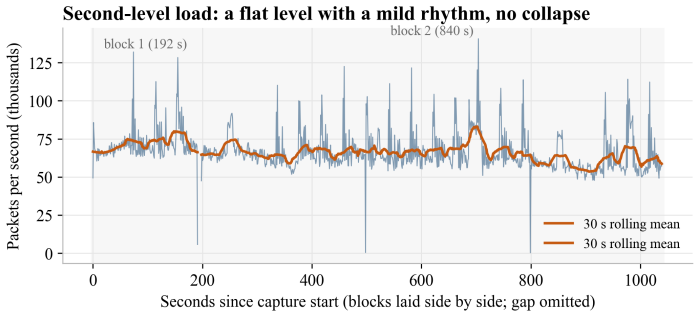
Figure 6: Outlier scatter: peers against ports, with the single host that falls off the chart

How to read this figure: both axes are logarithmic and the colour of each point indicates the group assigned by the transparent rule. The host worth noticing is 10.168.2.97, the outlier in the top-right corner whose peer count and port count both reach the measurement ceiling of 5,000 and which clearly stands apart from the rest of its group. It can only be treated as a lead, because a single host with no repeat sample cannot be distinguished from a host that is simply designed that way.

Whether the rule is more than a convenient label can be tested against a data-driven grouping. Running k-means on the behavioural features of the 5,223 inside hosts gives a silhouette (a measure of how compact a cluster is internally and how well separated it is from the others, closer to 1 being better) of 0.562 at k=4 and 0.565 at k=5 [6], a difference of 0.003, so k=4 is taken on grounds of parsimony, and the adjusted Rand index (a measure of agreement between two groupings, 0 being chance level and 1 a perfect match) against the rule groups is 0.41 [7], a moderate agreement. The cross-tabulation shows the direction of agreement, since 465 of the 478 serving hosts fall into one cluster and low-volume hosts likewise concentrate in one cluster, while the disagreement comes mainly from interactive hosts, which split roughly evenly across two clusters. Host behaviour therefore forms a continuum, and the rule is a transparent approximation of it rather than the only valid partition. This matters operationally: interactive hosts represent a behavioural continuum rather than a sharp group boundary, so manual monitoring will always meet borderline hosts, and thresholds and alert rules need headroom for cases that are not black and white instead of expecting every host to fall cleanly into one group.

4.4 Time dimension: stable, but only within these 17.2 minutes

Within the observation window the overall load is stable (see Figure 7). On the corrected two-block basis the first block (192 s) averages 71,735 packets per second with a coefficient of variation (the standard deviation divided by the mean, expressing volatility relative to level) of 0.163, and the second (842 s) averages 65,230 with 0.186, so neither shows a change of magnitude. The series also contains a rhythm that can be tested: the periodogram (the power spectrum of the series across candidate periods) [8] gives the first block a dominant period of 21.3 s with a peak 14.5 times the median of the shuffled baseline (permutation p = 0.022), and the second block's is 40.1 s at 92.2 times (p = 0.0025).

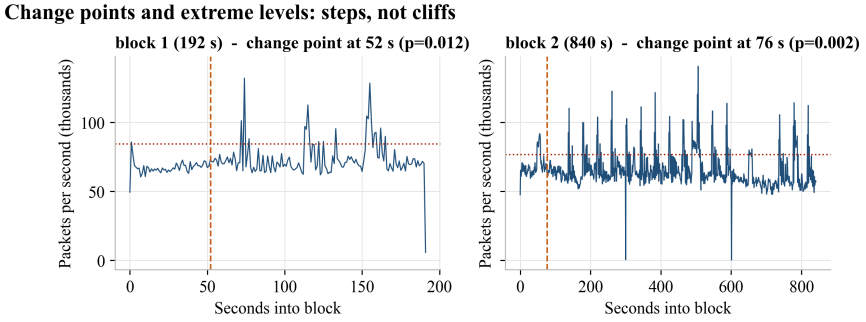


Coefficient of variation: block CV=0.163, block CV=0.186. No scaling event in either block. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 7: Second-level load: a stable platform with a mild rhythm

How to read this figure: the two blocks are shown side by side and the uncaptured interval between them is omitted, so the horizontal axis is not continuous time. The thin light line is the raw value of each individual second and the thick dark line is a 30-second rolling mean. The flatness of the rolling mean is the key detail, since it shows that no capacity event occurred inside this window and that any claim of an upward trend would require a longer dataset to support it.

Statistically significant change points and extreme levels [9] support the same reading (see Figure 8). Each block contains one dominant step, at second 52 and second 598, and their effects are 9.5% and 10.2% in the mean; analysed file by file instead, four steps of the same size appear. The shape parameter of the extreme-value fit is negative in all four files, so the upper tail is bounded, and the ten-second return level falls between 121,000 and 144,000 packets per second.



One significant change point per block (permutation p = 0.003-0.012), each moving the mean by at most about 11%. Extreme threshold (dotted line): block 1 84k pps (bounded tail, xi=-0.094); block 2 77k pps (bounded tail, xi=-0.196) - the negative shape parameter means a bounded tail. Source: cap_20260920_111416/112321/112821/113321.pcap (4 files, 17.2 min; two continuous blocks of 192 s and 842 s). One-way byte convention; loopback and non-IP traffic are handled per the data-convention note. Time-based statements are limited to those two blocks.

Figure 8: Change points and extreme levels: steps rather than cliffs

How to read this figure: each panel corresponds to one observation block, the dashed line marks the statistically significant change point and the dotted line marks the threshold used for the extreme-value analysis. The distinction between the two lines is what matters: the change point describes the mean stepping to a different level by no more than about 11%, while the negative shape parameter of the extreme-value fit means the upper tail is bounded. Together they show that the variation inside the window is a rhythm rather than an anomaly.

Seen across the whole window the residual band tells the same story. Once the trend is removed the second-level load stays inside the band for both blocks, with no interval in which the band suddenly widens, and the step between the blocks is only the mean moving from 71,735 to 65,230 packets per second, a shift within the same platform. The periodogram detail has to be read together with the shuffled baseline, because the second block's peak ratios are far higher than the first block's mainly because its series is longer and its random baseline flatter, not because the rhythm itself is stronger.

4.5 Outward identity and who initiates

At the level of names, the outward activity of this network looks like that of a typical office network. The most frequent names belong to device discovery (SSDP, mDNS, wpad) and to office collaboration (WeChat Work and similar), followed

by cloud services and telemetry and a small number of literal IP endpoints. The single most frequent entry is a multicast address used for device discovery rather than any service, which shows that a substantial part of the highest-ranked traffic is discovery and probing noise.

The distribution of session initiators explains where the directional asymmetry comes from. Inside hosts start 79.5% of all sessions, but sessions started from outside carry on average about twice as many bytes per session as those started from inside, and loopback sessions carry 1.7 MB each.

Host-level periodicity deserves a separate note, because its candidate set is large and its criterion loose. Of 1,461 host-by-block candidates, filtering to short periods with a clear peak leaves 1,039 rows, of which 59 exceed the shuffled baseline at the 5% level and only 29 reach five times it. A stricter route starts from 1,761 candidates and applies BH-FDR correction [10] (the Benjamini–Hochberg procedure, which controls the false discovery rate across multiple tests, a smaller q being stronger), leaving 1,096 significant at q below 0.05, of which 698 reach three times the baseline. The two conventions differ so widely because shuffling is a weak null, as it preserves the distribution of values while destroying their order, so any host with a mild rhythm looks significant. The resulting judgement is that second-level periodicity is common in an office network, since heartbeats, polling and keep-alive timers all produce it, and that it becomes an actionable lead only when a low coefficient of variation, a high peak and a clear contrast against comparable hosts hold at the same time.

4.6 Anomaly list

The four detections produced a number of candidates in directions that matter, summarised in the table below. It should be emphasised that every entry may only be written as a lead or as pending verification.

#	Phenomenon	Evidence (with denominator)	False-positive assessment	Status
A1	High fan-out host	10.168.2.97 hits the 5,000 ceiling on both peers and ports	one host, no repeat sample; may simply be designed that way	lead
A2	High-entropy wide-port hosts	159 hosts, 3.0% of the 5,223 inside hosts	tail of the serving group rather than an isolated case	lead
A3	Three fixed port groups	about 810,000 packets each, within 4 packets of one another	reads as parallel channels of one application	lead
A4	Outbound port monopolised by one external peer	three outbound ports each have 100% of their peers at a single public IP	no malicious features; needs business confirmation	lead
A5	Host-level periodicity	of 1,761 candidates, 1,096 significant after BH-FDR, 698 with a peak of at least 3x baseline	shuffling is a weak null; filter by low variation plus high peak	to verify
A6	Long-lived sessions	98,820 sessions lasting 60 to 300 s, 14.3% of 693,484	overlaps with elephant flows; needs cross-checking	to verify
A7	DNS ANY queries	6,404, or 1.66% of queries, counting name appearances	no amplification footprint, and the convention needs query-side calibration	lead
A8	DNS failures	NXDOMAIN at 8.3% of responses (9,770 of 117,463)	not unusual in a mixed office network	lead
A9	TLS handshakes without SNI	4,811 ClientHellos	consistent with the share of older clients	lead
A10	Change points	one significant step per block, 9.5% and 10.2% in the mean	small and present in both blocks, so more like a normal rhythm	lead

The rule for promotion should be stated explicitly: turning a lead into an anomaly requires all three of the conditions below, and at present no entry satisfies all three.

- **Statistical significance**: the difference must fall outside the range of random fluctuation.

- **A false-positive assessment**: it must be possible to describe what normal behaviour looks like for comparable hosts.
- **Comparison against normal behaviour**: a peer group is needed as a reference instead of a single observation.

5 Discussion and recommendations

For capacity planning the report supports three recommendations.

- **Tier the monitoring thresholds by group**: a handful of hosts have pulled the network-wide average up, so a single threshold produces a core working set that alerts continuously while the silent majority never alerts. Trade-off: more configuration items, since each group needs two or three tiers, and the grouping has to be revisited as host behaviour evolves.
- **Weight capacity assessment towards the inbound path and elephant flows**: bytes are 3.64 times larger inbound and the top hundred sessions carry a third of all session bytes. Trade-off: the assessment convention has to change, so earlier figures are not directly comparable with the new ones.
- **Plan connection-count and bandwidth capacity separately**: different factors dominate them, and mixing the two produces a wrong capacity figure. Trade-off: two capacity models have to be maintained, which raises the short-term assessment workload.

For security monitoring the report recommends the following.

- **Turn the four signals into alert rules**: fan-out, anomalous port entropy, bandwidth jumps and periodic heartbeats, each with a false-positive baseline and an FDR convention. Trade-off: false positives will be high at first, so a tuning period and a named reviewer are needed.
- **Write promotion conditions for every candidate**: this makes the step from lead to anomaly auditable rather than a matter of individual judgement. Trade-off: every candidate needs an owner and a criterion, which is a process cost.

Comparison against current practice has to be made carefully, because this report did not survey the monitoring and capacity practice actually in use, so what follows is conditional. If thresholds are currently uniform across all hosts, the concentration results in Section 4.2 imply that such a threshold is either high enough to leave the top hundred hosts permanently silent or low enough to keep the silent majority alerting, and if capacity is assessed by session or connection counts, the results in Section 4.5 imply a systematic underestimate of inbound bandwidth pressure, because bytes concentrate in a few elephant flows. Turning these statements from conditions into conclusions requires three answers from the operations side: how current thresholds are set, who reviews alerts, and which metric capacity planning uses.

Ordered by strength of evidence, the recommendations should land as follows: tier the thresholds first (strongest support, smallest change), separate the capacity metrics second (a change of convention rather than of equipment), and build the four alert rules last (a false-positive baseline has to accumulate first, so it takes the longest).

6 Improvements and next steps

Section 5 answered what can be done now; this section answers how the next measurement and analysis should be done. The goal is specific: to turn the candidates flagged in Section 4.6, and the conditional comparisons in Section 5, into decidable conclusions. Every recommendation is tied to a number in this report and states its cost.

6.1 Collection: making these 17.2 minutes comparable

- **Cover a full business cycle**: the observation window is two continuous blocks of 17.2 minutes in total (192 s + 842 s), so every timing statement in Section 4.4 is limited to that window. Later captures should cover at least a weekday peak, a night period and a weekend period at 1-second resolution; the cost is a retention policy decided up front, because the 36.2 GB captured in 17.2 minutes scales linearly and a full day places new demands on both storage and parsing.
- **Export device-side counters for cross-checking**: this report counts one-way bytes, each packet once, so its gigabyte figures cannot be compared directly with the bidirectional counters a device displays; this is the convention most likely to be challenged. Taking SNMP interface counters or NetFlow records over the same window would let the ratio and the residual validate the one-way convention; the cost is device access, export configuration and clock alignment between the two sides.
- **Replace the three-tuple approximation with connection tracking**: sessions here are approximated by (source, destination, destination port) (693,484 of them) with no start or end marker, so the session results in Section 4.5 are approximate. Enabling flow export or connection tracking would make session duration and elephant flows directly measurable rather than inferred; the cost is additional volume and parsing work.

6.2 Analysis: turning candidates into decisions

- **Re-run the periodicity test with surrogates that keep short-range dependence**: the permutation test uses random shuffling as its null, and shuffling destroys the short-range dependence of bursty traffic, which makes it a weak null,

as the methodology notes. A surrogate method that preserves autocorrelation, such as block resampling or phase randomisation, would firm up the periodicity result; the cost is one round of method work and review.
- **Give the grouping an external, business-side reference**: the four groups from the transparent rule agree only moderately with the data-driven k-means partition (ARI = 0.41), so the groups are still a statistical cut. Adding asset-database or name-resolution data as an external reference would turn the grouping from a statistical construct into a business fact; the cost is a dependence on other systems' availability and consistency.
- **Report effect sizes alongside intervals**: significance is already controlled by the Benjamini-Hochberg procedure (1,096 significant and 665 not significant at the extreme-value threshold), but significance is not the same as a meaningful magnitude. Later statements should carry both a magnitude and an uncertainty interval; the cost is a longer report and more charts.

6.3 Delivery: from a one-off report to a routine check-up

- **Fix the metrics and conventions into a single check-up sheet**: this report uses six metrics, namely the direction ratio, the top-N% concentration, the coefficient of variation, the dominant period, port entropy and the clustering agreement. Fixing them under one convention, writing that convention into the operations documentation and reporting monthly would make the numbers comparable across time; the cost is scheduling and retention, though the figure scripts already rebuild all 24 figures with one command inside the repository (see Section 9).
- **Build a baseline before talking about change**: with no historical baseline, "stable" can only describe these 17.2 minutes. Accumulating about three months of per-group quantile baselines, and hanging the candidate list's criteria on them, would turn the leads into measurable change; the cost is three months of continuous running with no change of convention in between.

Recommendation	Unlocks	Cost	Priority
Fix the metrics into a check-up sheet	"stable" gains a comparable reference	very low, no new data	immediate
Cover a full business cycle	from "no event in the window" to "peak period of that day"	storage and retention	high
Device-side counter cross-check	removes the one-way challenge	device access and clock alignment	high
Monthly reruns and a baseline	candidates become comparable change	about three months of running	medium
Surrogate-based periodicity test	periodicity stable beyond one window	method work	medium
Connection tracking instead of three tuples	session results measurable, not inferred	volume and parsing	medium

7 Limitations

The report has three groups of limitations, all of which must be kept in mind when its conclusions are quoted.

Capture-side limitations

- **A very short window**: 17.2 minutes on one morning in two continuous blocks, so no claim is made about daily or weekly patterns.
- **Segment character rests on sampling**: the sampled window covers the first 500,000 packets of each file rather than a random interval, and only connections that asked for or presented a name.
- **Privacy convention**: categories only (instant messaging, office suites, print discovery, remote desktop, telemetry, network devices); no host names, device models or personal domains.

Statistical limitations

- **DNS metric convention**: name appearances are counted with query and response side each counted once, so individual shares require query-side calibration before being quoted.
- **A weak null for the periodicity test**: the host-level test uses a shuffled series as its null hypothesis, which is not equivalent to confirming a business rhythm.
- **The grouping rule is analyst-chosen**: its thresholds (byte P90, 50 ports, 20 peers) were selected by the analyst and agree only moderately with a data-driven clustering (ARI 0.41), so group boundaries carry a conventional element.

Interpretation-side limitations

- **Unconfirmed candidates**: the candidates in Section 4.6 have not been confirmed by the business side and remain leads only.

- **Current practice was not surveyed**: the comparison in Section 5 is conditional, and three answers from the operations side are needed before it can be turned into conclusions.

8 Conclusions

This chapter answers the three research questions of Section 1 in turn, with the location of the evidence for each answer in brackets.

Addressing RQ1 (the structure of this network): this is an aggregation network that is receive-dominant, in which a few core hosts move most of the data. Averages fail here, since the top 1% of hosts carry 81.6% of the bytes and the top hundred alone carry 59.9%, so management has to be tiered. Inside hosts take four roles, namely serving, interactive, bulk and low-volume, and 478 serving hosts carry 72.2% of the inside bytes. The network presents an encrypted service face to the outside, with outbound packets on port 443 running 5,317 times the inbound count, and a service mesh on the inside. The external structure is uneven as well, since the top 1% of /24 blocks carry 85.0% of the external bytes (evidence in Sections 4.1 to 4.3 and 4.5).

Addressing RQ2 (stability in time): inside the 17.2-minute window the network is stable, with coefficients of variation of 0.163 and 0.186 in the two blocks and no change of capacity significance, and it shows a testable rhythm, with dominant periods of 21.3 s and 40.1 s, change points of only 9.5% and 10.2%, and a bounded upper tail. The variation inside the window is therefore a rhythm rather than an anomaly. The two classes of finding do not carry the same weight, however: the same structure recurs in all four capture files, whereas the time patterns are limited by the length of the window and cannot be extrapolated (evidence in Section 4.4).

Addressing RQ3 (leads worth following): the four detections produced ten candidates, and none of them satisfies all three of statistical significance, a false-positive assessment and a comparison against comparable hosts, so they remain leads rather than conclusions. Host-level periodicity needs particular care here, because shuffling is a weak null that makes many ordinary hosts look significant (evidence in Sections 4.5 and 4.6). The operational implication is that security alerts should not be triggered by statistical periodicity alone, without host-group context such as whether comparable hosts in the same group show the same rhythm.

9 Reproducibility and toolkit

The analysis code, the derived data and the scripts that generate all 24 figures are kept in a single repository, which reproduces the whole figure set when cloned; the raw captures are not redistributed. Repository: **ChiantiRuan/ST5290-Campus-Traffic-Analysis**

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